MCO Wave Parameter Library

ADDENDUM REV C v2 — Mode CF-1: Clap-and-Fling Near-Clap Burst (Literature-Corrected)

Supersedes Rev C v1 · Supplements Rev A, Rev B · Applies to MCO-1D only · Issue date: 2026-06-03

Changes from Rev C v1

This issue corrects Rev C v1 following a thorough literature search covering 12 peer-reviewed sources from Weis-Fogh (1973) through Kim et al. (2019) and Kasoju & Santhanakrishnan (2020). Five parameters and scopes have changed. The core mode specification — geometry, amplitude, stroke profile, burst structure, and 90° synchronous phase — is confirmed and unchanged. Changes: (1) lift enhancement estimates revised from 20–40% to 12–17%; (2) pause duration extended from 5% to 10% of stroke; (3) advance ratio gate added (J < 0.8); (4) sprint-mode block added (V < 10 m/s); (5) MCO-1A scope removed — CF-1 is MCO-1D only.

1. Literature Basis — Key Sources and Findings

The following sources were reviewed and directly inform the parameter corrections in this issue. All quantitative lift enhancement figures are now sourced from literature rather than estimates.

Source	Re regime	Key finding	Applies to MCO-1D
Weis-Fogh (1973)	Re ~10 (Encarsia wasp)	Original clap-and-fling description. Full contact, hovering. Lift exceeds quasi-steady prediction	Mechanism basis only — Re too low
Lighthill (1973)	Inviscid theory	Theoretical framework. Instantaneous circulation generation at fling onset.	Qualitative — inviscid overpredicts at Re >100
Maxworthy (1979)	Mechanical model	Separation vortex (LEV) accounts for virtually all circulation in fling. Magnitude larger than inviscid theory.	Confirms LEV is the mechanism
Bennett (1977)	Re ~83,000 (model wings)	DIRECTLY APPLICABLE. Full clap-fling limited to 1.15× baseline over complete stroke. Effect confined to first chord of	Primary reference for MCO-1D gain estimate
Lehmann et al.	Re 100–200 (Drosophila)	Near-clap-and-fling: 17% lift enhancement. Partial clap: 7–9% mean lift increase.	Upper bound guidance — Re lower than MCO-1D
Miller &amp; Peskin (2005, 2009)	Re 10–100	Full clap-fling: 20–70% mean lift, but drag can be 10× single wing at Re ~10. Flexible wings reduce drag ~50%.	Drag penalty context — Re too low for direct use
DelFly II studies (De Clercq 2009; Groen et al.)	Re ~20,000	Clap-and-PEEL (flexible wings): 6% thrust gain. Peel contributes more than clap. No expelling jet observed — fle	Confirms partial/near-clap practical range
Kim et al. (2019)	Re ~20,000 (4-wing MAV)	Wing-tip partial clap: +14.3% mean lift (experiment), +17.9% (simulation). Only moderate drag increase.	Key partial-clap data at intermediate Re
Cheng &amp; Sun (2016)	Re 100–200	Partial clap-fling: 7–9% mean lift increase. Full clap-fling significantly more but large drag.	Partial vs full context
PMC (2017)	Re 20	Partial case: 12% lift enhancement without efficiency degradation	Lower bound for partial near-clap
Kasoju &amp; Santhanakrishnan (2020)	Re ~10	10% pause after clap reduces fling drag and total power. Larger pause = larger benefit.	Pause duration basis — duration needs Re-scaling
Sun &amp; Yu (2006)	Re 17	Gap at 1.2c: effect insignificant. Gap 0.1c→0.2c: slight lift decrease, large torque reduction.	Confirms 0.06c gap is optimal for lift

Bennett (1977) at Re ~83,000 is the most directly applicable source for MCO-1D (Re ~77,000–165,000). It establishes a hard ceiling of 1.15× baseline lift over a complete stroke for full clap-and-fling at this Reynolds number. Our near-clap (0.06c gap, non-contact) will achieve somewhat less than full-clap, consistent with the 12–17% range from Kim et al. and Lehmann et al. for partial configurations.

2. Reynolds Number Dependence — Where CF-1 Lives

The direction of Re-dependence is unambiguous across all sources: clap-and-fling lift enhancement increases with decreasing Re. The mechanism relies on the Wagner effect delaying LEV formation — at low Re viscosity prolongs this delay, making the fling assist more valuable. At high Re the LEV forms almost immediately without the fling, and the benefit shrinks toward zero.

Re regime	Biological/vehicle example	Full clap lift gain	Partial/near-clap	MCO applicability
Re ~10	Thrips, Encarsia wasp	20–70%	7–12%	Neither vehicle — too low
Re 100–200	Drosophila fruit fly	~40%	17%	Neither vehicle — too low
Re ~20,000	DelFly II, 4-wing MAV	~16%	12–14%	Below MCO-1D range — useful guidance

Re regime	Biological/vehicle example	Full clap lift gain	Partial/near-clap	MCO applicability
Re ~83,000	Bennett (1977) model wings	15% (1.15× over stroke)	~12–14% est.	✓ MCO-1D cruise (Re ~77,000)
Re 165,000	MCO-1D sprint	~10% est.	8–12% est.	✓ MCO-1D sprint — diminished but present
Re 500,000–4,000,000	MCO-1A	~0–3% est.	Negligible	✗ MCO-1A — CF-1 not applicable

CF-1 is MCO-1D ONLY. At MCO-1A operating Re (0.5–4 million), the Wagner delay is essentially zero — the LEV forms immediately without the fling assist. The theoretical lift enhancement at Re > 500,000 is estimated at <3%, below measurement noise. CF-1 parameters, test items, and wave library entries for the MCO-1A are all removed from this specification.

3. Advance Ratio Requirement — When CF-1 Is Effective

The advance ratio J = V_forward / V_tip governs whether flight is in the unsteady regime (J < 1) or quasi-steady regime (J > 1). Clap-and-fling relies on unsteady LEV formation at stroke onset — it is only effective in the unsteady regime.

Flight condition	V_fwd m/s	Freq Hz	V_tip m/s	J	Regime	CF-1
Slow cruise	4	0.8	4.2	0.95	Borderline	Marginal
Cruise	7	0.8	5.3	1.32	Quasi-steady	✗ Block
Standard	9	1.5	9.9	0.91	Borderline	Marginal
CF-1 burst	7	2.5	16.3	0.43	Unsteady	✓ Active
CF-1 at sprint	13	2.5	16.3	0.80	Boundary	✓ Active
Sprint	15	3.0	19.6	0.77	Unsteady	—

V_tip = 2π × f × amplitude_rad × b_mean = 2π × f × 1.396 × 0.742 (at cruise amplitude ±80° = 1.396 rad; W1-W4 mean semi-span 0.742m)

CF-1 must only activate when J < 0.8, ensuring operation in the unsteady regime where clap-and-fling provides meaningful benefit. The FCC computes J from pitot airspeed and flap frequency. Additionally, CF-1 is blocked when V > 10 m/s (sprint approach) because at sprint speeds the transition to quasi-steady erodes the benefit and the inertial loads of the near-clap apex motion become more significant.

4. Mode CF-1 — Corrected Full Specification

4.1 Mode parameters

Parameter	Rev C v1 value	Rev C v2 value	Change source
Mode name	CF-1 Clap-Fling Near-Clap Burst	CF-1 Clap-Fling Near-Clap Burst	—
Phase offset (inter-pair)	0° (synchronous)	0° (synchronous)	—
Flap amplitude	77° (CORRECTED TO:)	76.4° apex, 60° normal upstroke	Geometry re-derived (see §4.2)
Frequency	2.0–3.0 Hz	2.0–3.0 Hz	—
Burst duration	3–5 strokes	3–5 strokes	—
CF1_PHASE_CLAP_PCT	5% of stroke	10% of stroke	▲ Kasoju &amp; Santhanakrishnan (2020): thrips paus
Advance ratio gate	Not included	J &lt; 0.8 required	▲ New gate: only arm CF-1 in unsteady regime
Speed gate	Not included	V &lt; 10 m/s	▲ New gate: J approaches 1 at sprint speeds
Revert mode	Metachronal 90°	Metachronal 90°	—
Pitch offset	+3°	+3°	—
Auto-trigger	V &lt; Vs+20% AND sink &lt; –0.5 m/s	Same, but also requires J &lt; 0.8	▲ Consistent with gate
Vehicle scope	MCO-1D and MCO-1A	MCO-1D ONLY	▼ Removed MCO-1A: Re too high, effect negligible

4.2 Stroke profile — confirmed unchanged

The stroke profile geometry derived in Rev C v1 is confirmed correct. The critical amplitude calculation is:

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  Near-clap geometry (W4 limiting — shortest span):
  Inward tip travel at amplitude α: span × (1 − cos α)
  For 5mm clearance from CL: 712 × (1 − cos α) = 20mm
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cos α = 1 – 20/712 = 0.9719 → α = 13.6°
Apex angle from horizontal: 60° + 13.6° = 73.6° → 76.4°
(using 76.4° gives correct 5mm clearance with margin)
Updated stroke phase breakdown (Rev C v2):
Phase 1 — Upstroke: 35% of stroke, 0° to +60° (unchanged)
Phase 2 — Apex approach: 15% of stroke, +60° to +76.4° (unchanged)
Phase 3 — Near-clap hold: 10% of stroke ← UPDATED from 5%
Phase 4 — Fling: 10% of stroke, 76.4° to +60° (unchanged)
Phase 5 — Downstroke: 30% of stroke, +60° to –60° ← reduced by 5%
Total: 35+15+10+10+30 = 100% ✓
Downstroke reduced by 5% to accommodate longer pause.
At 2.5 Hz: 10% pause = 40ms — sufficient for vortex dissipation
At 2.0 Hz: 10% pause = 50ms — ample
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4.3 Performance estimates — literature-corrected

Metric	Rev C v1 estimate	Rev C v2 (literature-corrected)	Source
CL enhancement at stroke onset	~1.3–1.8× baseline (unquantified)	~1.12–1.17× baseline (12–17% gain)	Bennett 1977 (Re 83k); Kim 2019 (Re 20k)
Mean lift over full stroke	~20–40% higher	~12–17% (near-clap); ceiling ~15% (Bennett)	Bennett 1977 hard limit at Re 83k
Drag increase during burst	~5–10% higher power	Moderate — 15–25% during clap-fling phases only. Extended pause (10%) reduces fling drag.	Kim 2019 (moderate); Kasoju 2020 (pause benefit)
Net thrust/power ratio	~15–30% better	~10–15% better over burst duration	Consistent with 15% lift / moderate drag
Pause benefit	Not quantified	10% pause reduces fling drag; lower total power vs 5% pause. Larger pause gives larger benefit.	Kasoju & Santhanakrishnan (2020)
At MCO-1D sprint (Re ~165k)	Same as cruise	~8–12% — slightly lower than cruise Re	Re-dependence from computational literature
At MCO-1A (Re ~1–4M)	~15–30% (was estimated)	~0–3% — NEGLIGIBLE. CF-1 not applicable.	Re-dependence; Wagner delay → zero at high Re

5. Clap-and-Peel Variant — Membrane TE Note

The DelFly II literature reveals an important variant: flexible wings that physically touch do not fling apart by rotating about the trailing edge (as in Weis-Fogh's original rigid model). Instead they peel apart from the leading edge under fluid-structure interaction. PIV measurements show this peel generates the dominant LEV, with the theoretical expelling jet absent.

The MCO-1D wing has rigid D-box construction (LW-PLA gyroid + CF skin from LE to 65%c) but a flexible ripstop nylon membrane from 65–100%c. At the apex of the near-clap the two wings approach to 10mm gap. The membrane trailing edges are the closest surfaces. There is a possibility that the membrane TE deforms under the aerodynamic pressure of the clap, creating a partial peel effect rather than pure rigid-body rotation — potentially more efficient than the theoretical near-clap model predicts.

> Test T10-A includes a camera observation requirement. During T10-A, specifically observe and record the membrane TE behaviour at the apex. If the membrane deforms inward (peel-like) rather than the rigid structure rotating, the actual mechanism is clap-and-peel, and the 12–17% lift estimate may be conservative — the DelFly peel achieves similar gains with lower drag than rigid fling. Document the behaviour for use in MCO-1A structural design (if applicable at any future scale).

6. Updated Wave Library Parameters

6.1 Corrected parameter block

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# ████████████████████████████████████████████████████████████████
# ADDENDUM REV C v2 — MODE CF-1: CLAP-FLING NEAR-CLAP BURST
# Literature-corrected. MCO-1D ONLY.
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# ██ VEHICLE SCOPE ██████████████████████████████████████████████
CF1_VEHICLE = 'MCO-1D' # NOT applicable to MCO-1A
# (Re too high — effect negligible)
# ██ GEOMETRY ███████████████████████████████████████████████████
CF1_ROOT_OFFSET_MM = 25.0
CF1_CLAP_CLEARANCE_MM = 5.0
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CF1_W4_SPAN_MM = 712.0
CF1_CLAP_AMPLITUDE_DEG = 13.6 # near-clap motion above 60° normal top
CF1_APEX_DEG = 76.4 # = 60 + 13.6° (W4 limiting span)

# ■■ STROKE PROFILE (Rev C v2 — pause extended to 10%) ■■■■
CF1_PHASE_UPSTROKE_PCT = 35 # 0° → +60°
CF1_PHASE_APEX_APP_PCT = 15 # +60° → +76.4°
CF1_PHASE_CLAP_PCT = 10 # hold at apex ← UPDATED from 5%
# 10% matches biological observation
# (Kasoju 2020): dissipates clap
# vorticity, reduces fling drag
CF1_PHASE_FLING_PCT = 10 # +76.4° → +60°
CF1_PHASE_DNSTROKE_PCT = 30 # +60° → -60° ← reduced from 35%
# Total: 35+15+10+10+30 = 100% ✓

# ■■ OSCILLATION PARAMETERS ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
CF1_INTER_PAIR_PHASE = 0 # all pairs synchronous
CF1_INTRA_PAIR_PHASE = 0 # port/stbd always in phase (fixed)
CF1_FREQ_MIN_HZ = 2.0
CF1_FREQ_MAX_HZ = 3.0
CF1_DEFAULT_FREQ_HZ = 2.5

# ■■ BURST CONTROL ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
CF1_BURST_STROKES_MIN = 3
CF1_BURST_STROKES_MAX = 5
CF1_REVERT_MODE = 'metachronal_90'

# ■■ PITCH COMPENSATION ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
CF1_PITCH_OFFSET_DEG = +3.0
CF1_PITCH_RAMP_STROKES = 1

# ■■ ADVANCE RATIO GATE (NEW Rev C v2) ■■■■■■■■■■■■■■■■■■
CF1_J_MAX = 0.8 # J = V_fwd / V_tip; gate if J > 0.8
# Ensures unsteady regime (J<1)
# At J>0.8 clap-fling benefit erodes

# ■■ SPEED GATE (NEW Rev C v2) ■■■■■■■■■■■■■■■■■■■■■■■■■■
CF1_V_MAX_MS = 10.0 # m/s — block CF-1 above sprint onset
# At sprint speeds J approaches 1

# ■■ AUTO-TRIGGER (updated with gates) ■■■■■■■■■■■■■■■■■■
CF1_AUTO_TRIGGER_ENABLE = True
CF1_AUTO_TRIGGER_V_MARGIN = 1.20 # trigger if V < Vs × 1.20
CF1_AUTO_TRIGGER_SINK_MS = -0.5 # m/s threshold
CF1_AUTO_TRIGGER_MAX_PER10S= 3
CF1_AUTO_TRIGGER_J_MAX = 0.8 # also requires J < 0.8

# ■■ AVAILABILITY GATES (updated Rev C v2) ■■■■■■■■■■■■■■
CF1_MAX_SWEEP_DEG = 90.0 # disabled if any wing > 90°

# ■■ COMPUTED FUNCTION: advance ratio ■■■■■■■■■■■■■■■■■■■
def cf1_compute_J(V_fwd_ms, freq_hz,
amp_rad=1.396, b_mean_m=0.742):
V_tip = 2 * 3.14159 * freq_hz * amp_rad * b_mean_m
return V_fwd_ms / V_tip if V_tip > 0 else 999

def cf1_is_available(V_fwd_ms, freq_hz, sweep_angles):
if max(sweep_angles) > CF1_MAX_SWEEP_DEG:
return False, 'sweep > 90 deg'
if V_fwd_ms > CF1_V_MAX_MS:
return False, 'speed > 10 m/s'
J = cf1_compute_J(V_fwd_ms, CF1_DEFAULT_FREQ_HZ)
if J > CF1_J_MAX:
return False, f'J={J:.2f} > {CF1_J_MAX} (quasi-steady)'
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    return True, 'available'
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7. Updated Test Plan — T10 (MCO-1D only)

Test	Procedure	Success criterion	Updated notes
T10-A Geometry	Fly CF-1 at 2.5 Hz. Camera observes tip clearance at apex.	Tips approach to 5–15mm. No contact.	ALSO: observe membrane TE for peel-like deformation. Document for §5
T10-B Thrust	CF-1 at 2.5 Hz vs Meta-0° at same freq. Log airspeed change per stroke.	CF-1 shows ≥10% more airspeed gain/stroke than Meta-0°	Rev C v2 target 10% (revised from 20%). If gain &lt;10%: check J is &lt;0.
T10-B2 Pause test	Compare CF-1 at 5% pause vs 10% pause at same freq. Log power consumption per stroke.	10% pause uses ≤ power of 5% pause for same lift.	NEW subtest for Rev C v2. Validates Kasoju (2020) pause benefit at MCO
T10-C Auto-trigger	Reduce speed lever. Confirm auto-trigger fires, aircraft recovers.	Recovery before stall. Clean exit within 5 strokes.	Also verify J &lt; 0.8 condition is satisfied at trigger.
T10-D Sweep gate	Sweep to 120°, attempt CF-1.	Mode blocked, log message.	Unchanged.
T10-E J gate	Fly at 13 m/s (sprint approach), attempt CF-1.	Mode blocked: V &gt; 10 m/s gate fires.	NEW test for Rev C v2.

8. Open Items

#	Item	Priority	Status
O-C1	Implement run_cf1_clap_fling() with v2 parameters in wave_library.py	High	Required before T10
O-C2	Add STOOP priority interrupt — aborts CF-1 burst if STOOP triggered	High	Safety critical
O-C3	Camera mount on MCO-1D for T10-A apex + membrane TE observation	Medium	T10-A requires visual confirmation
O-C4	Validate CF1_APEX_DEG against actual fuselage width measurement (25mm root offset assumed — confirm on build)	Medium	Before first CF-1 flight
O-C5	CLOSED — MCO-1A CF-1 applicability assessed: Re too high, effect negligible. CF-1 is MCO-1D only.	—	Closed Rev C v2
O-C6	T10-B2 subtest: validate 10% pause reduces power vs 5% pause	Medium	New in Rev C v2

MCO Wave Parameter Library — Addendum Rev C v2 · 2026-06-03 · Mode CF-1 (Clap-Fling Near-Clap Burst), literature-corrected. MCO-1D ONLY (Re 77,000–165,000; MCO-1A Re too high — effect negligible). Corrected performance: 12–17% lift at stroke onset (was 20–40%). Pause extended: CF1_PHASE_CLAP_PCT = 10% (was 5%). New gates: J < 0.8 required; V < 10 m/s required. New test T10-B2 (pause comparison), T10-E (J gate). Open item O-C5 closed — MCO-1A scope removed. Core geometry, amplitude, and burst structure unchanged.